Ans. 4 equally likely outcomes: $\frac{1}{4}$.

5. Thread · Method [ALG] Programme: choose a number; square it; add 1. Result for -3? Can 0 be obtained? Justify.

   Ans. For -3: $(-3)^2+1=10$. No: $x^2 \geqslant 0$ so $x^2+1 \geqslant 1 > 0$.

Sheet 29 Mon 11 Jan. 2027 *Grouped data: histogram and cumulative frequencies*

1. Warm-up · Quick [STA] Give the width and the midpoint of the class $[20;35[$.

   Ans. Width 15; midpoint $\frac{20+35}{2} = 27.5$.

2. Core · Table [LEC] Classes $[0;10[$, $[10;20[$, $[20;30[$ with frequencies (counts) 4, 6, 10. Give the relative frequencies, then the cumulative increasing relative frequencies.

   Ans. Total 20; relative frequencies 20 %, 30 %, 50 %; cumulative 20 %, 50 %, 100 %.

3. Core + · Quick [LEC] In a histogram with classes of equal width, two rectangles have heights 3 cm and 6 cm. What is the ratio of their frequencies (counts)?

   Ans. The heights are proportional to the counts: 1 to 2.

4. Review · Quick [NUM] Write 0.00032 in scientific notation.

   Ans. 3.2×10^{-4}.

5. Thread · Quick [EQ] Solve in $\mathbb{R}$: $2(x+3) = 5x - 9$.

   Ans. $x = 5$

   Pitfall: forgetting to distribute the 2 over the 3.

Sheet 30 Tue 12 Jan. 2027 *Weighted mean, median class; spreadsheet and Python*

1. Warm-up · Quick [CAL] Calculate $\frac{2\times 8+3\times 12}{5}$.

   Ans. $\frac{52}{5} = 10.4$.

2. Core · Method [STA] Classes $[0;10[$, $[10;20[$, $[20;30[$ with frequencies (counts) 4 ; 7 ; 9. Estimate the mean using the class midpoints; give the median class.

   Ans. Estimated mean $\frac{4\times 5+7\times 15+9\times 25}{20} = 17.5$; cumulative frequencies 4 ; 11 ; 20: the 10^{e} and 11^{e} values are in $[10;20[$: median class.

3. Core + · Code reading [ALG] What does this programme calculate and what value does it give?

```
notes = [12, 15, 9, 14]
s = 0
for n in notes:
    s = s + n
print(s / len(notes))
```

   Ans. The mean of the marks: $\frac{50}{4} = 12.5$.

4. Review · Quick [PRO] An item costing 45 000 Ar is sold at -30 %. What is the sale price?

   Ans. $45\,000 \times 0.70 = 31\,500$ Ar.

5. Thread · Prove [LIT] Programme: choose n; add 1; square it; subtract n^2. Show that the result is always an odd number.

   Ans. $(n+1)^2 - n^2 = 2n+1$: odd for every integer n.

Algebra, equations and inequalities

Sheet 31 Thu 14 Jan. 2027 *Integer powers (positive and negative)*

1. Warm-up · Quick [PUI] Calculate 2^{-3}, 10^{-2} and $(-3)^2$.

   Ans. $\frac{1}{8}$; 0.01; 9.

2. Core · Method [PUI] Simplify $\dfrac{2^3 \times 2^{-5}}{2^{-1}}$.

   Ans. $\dfrac{2^{-2}}{2^{-1}} = 2^{-2+1} = 2^{-1} = \dfrac{1}{2}$.

3. Core + · Method [PUI] For $x \neq 0$, write $(x^2)^{-3} \times x^7$ in the form x^n.

   Ans. $x^{-6} \times x^7 = x^1 = x$.

4. Review · Quick [STA] Median of the series 4, 7, 9, 10, 12, 15?

   Ans. Even number of values: $\dfrac{9+10}{2} = 9.5$.

5. Thread · Identify [SOL] Which solid has a disc-shaped base, a curved lateral surface and a single vertex?

   Ans. A right circular cone.

Sheet 32 Mon 18 Jan. 2027 *Square roots; $\sqrt{a^2} = |a|$*

1. Warm-up · Quick [CAL] Calculate $\sqrt{36}$ and $\sqrt{0.09}$.

   Ans. 6; 0.3.

2. Core · Method [CAL] Write $\sqrt{72}$ in the form $a\sqrt{2}$ and calculate $\sqrt{(-5)^2}$.

   Ans. $\sqrt{72} = \sqrt{36 \times 2} = 6\sqrt{2}$; $\sqrt{(-5)^2} = |-5| = 5$.

3. Core + · True/false [CAL] True or false? Justify. $\sqrt{9+16} = \sqrt{9} + \sqrt{16}$.

   Ans. False. $\sqrt{25} = 5$ but $3 + 4 = 7$. Only the product "splits": $\sqrt{ab} = \sqrt{a}\sqrt{b}$.

4. Review · Quick [PUI] Calculate $(3^2)^3$.

   Ans. $3^{2\times 3} = 3^6 = 729$.

5. Thread · True/false [DEM] True or false? Justify. The diagonals of a rhombus are perpendicular.

   Ans. True. Property of the rhombus (its diagonals are also perpendicular bisectors of each other).

Sheet 33 Tue 19 Jan. 2027 *Fractional expressions; expanded, reduced or factorised forms*

1. Warm-up · Quick [CAL] Calculate $\dfrac{1}{2} + \dfrac{1}{3}$.

   Ans. $\frac{5}{6}$.

2. Core · Method [LIT] Write as a single fraction $\dfrac{1}{x} + \dfrac{1}{x+1}$ (for $x \neq 0$ and $x \neq -1$).

   Ans. $\dfrac{x+1+x}{x(x+1)} = \dfrac{2x+1}{x(x+1)}$.

3. Core + · Method [DEV] Factorise $x^2 - 4$ then simplify $\dfrac{x^2-4}{x-2}$ for $x \neq 2$.

   Ans. $x^2 - 4 = (x-2)(x+2)$; so $\dfrac{x^2-4}{x-2} = x + 2$.

   The choice of form depends on the goal: factorised to simplify, expanded to compare.

4. Review · Quick [SUB] Calculate $2a - 3b$ for $a = 2$ and $b = -1$.

   Ans. $4 + 3 = 7$.

5. Thread · Quick [REP] $ABCDEFGH$ is a cuboid with $AB = 4$, $AD = 3$, $AE = 2$. In the frame $(A; B, D, E)$, what are the coordinates of G?

   Ans. $G(4; 3; 2)$: G is the vertex opposite A.

   A third coordinate appears in space.

Sheet 34 Thu 21 Jan. 2027 *Inequalities: sum, multiplication by a real number*

1. Warm-up · Quick [CMP] Complete with $<$ or $>$: $-3 \ldots -2$; $\frac{1}{3} \ldots 0.3$.

   Ans. $-3 < -2$; $\frac{1}{3} > 0.3$ ($0.333\ldots > 0.3$).

2. Core · Method [EQ] We have $a < b$. Compare $-2a$ and $-2b$, then $a+5$ and $b+5$.

   Ans. $-2a > -2b$ (multiplying by a negative number reverses the inequality); $a+5 < b+5$.

3. Core + · Method [EQ] Given $2 < x < 3$ and $1 < y < 4$, find bounds for $x+y$ then $x-y$.

   Ans. $3 < x+y < 7$; for $x-y$: $-y \in]-4;-1[$ so $-2 < x-y < 2$.

   Pitfall: subtracting the inequalities term by term.

4. Review · Quick [CAL] Calculate $(-3) \times (-4) + 2 \times (-5)$.

   Ans. $12 - 10 = 2$.

5. Thread · Problem [DIV] A 120 cm by 84 cm sheet is cut into identical squares, with no waste. What is the largest possible side length?

   Ans. PGCD(120 ; 84) $= 12$: squares of side 12 cm ($10 \times 7 = 70$ squares).

Sheet 35 Mon 25 Jan. 2027 *Comparing by difference and by quotient*

1. Warm-up · Quick [CMP] Compare $\frac{2}{3}$ and $\frac{3}{4}$ by studying the sign of the difference.

   Ans. $\frac{2}{3} - \frac{3}{4} = -\frac{1}{12} < 0$: $\frac{2}{3} < \frac{3}{4}$.

2. Core · Method [LIT] Compare x^2+1 and $2x$ for every real number x.

   Ans. $x^2 + 1 - 2x = (x-1)^2 \geqslant 0$: $x^2 + 1 \geqslant 2x$, with equality if and only if $x = 1$.

3. Core + · Method [CMP] Compare $\frac{7}{5}$ and $\frac{8}{6}$ by quotient (positive numbers).

   Ans. $\frac{7/5}{8/6} = \frac{42}{40} > 1$, so $\frac{7}{5} > \frac{8}{6}$.

4. Review · Table [LEC] Monthly rainfall (fictional data).

Month	Jan.	Feb.	Mar.	Apr.
Rain (mm)	280	300	240	120

   Which month is the wettest? What is the total over these 4 months?

   Ans. February (300 mm); 940 mm.

5. Thread · Table → formula [REG] x: $-2, -1, 0, 1$ and $f(x)$: $4, 1, 0, 1$. Is the function affine? Suggest a formula.

   Ans. No (the changes are not constant). $f(x) = x^2$ works.

   Test: $f(1) - f(0) = 1$ *but* $f(0) - f(-1) = -1$.

Sheet 36 Tue 26 Jan. 2027 *First-degree equations and inequalities; $x^2 = a$; zero product*

1. Warm-up · Quick [EQ] Solve $3x - 5 = 7$.

   Ans. $x = 4$.

2. Core · Method [EQ] Solve $2x + 3 \leqslant 7 - x$.

   Ans. $3x \leqslant 4$: $x \leqslant \frac{4}{3}$; $S = \left]-\infty; \frac{4}{3}\right]$.

3. Core + · Method [EQ] Solve $(x-2)(2x+6) = 0$ then $x^2 = 7$.

   Ans. Zero product: $x = 2$ or $x = -3$; $x^2 = 7$: $x = \sqrt{7}$ or $x = -\sqrt{7}$.

4. Review · Quick [CMP] Compare 1.4 and $\sqrt{2}$ without a calculator.

   Ans. $1.4^2 = 1.96 < 2$ so $1.4 < \sqrt{2}$ (the square function is increasing on $[0; +\infty[$).

5. Thread · Quick [GEO] ABC is isosceles at A with $\widehat{BAC} = 40°$. What is the measure of $\widehat{ABC}$?

   Ans. $(180 - 40) \div 2 = 70°$.